

exp 28.py - C:/Users/shaik/OneDrive/Desktop/MLA0405/exp 28.py (3.13.5)

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```
import matplotlib.pyplot as plt
from sklearn.datasets import make_blobs
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score
```

```
X, y = make_blobs(
    n_samples=500, centers=3, n_features=2,
    random_state=42
)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

```
model = MLPClassifier(
    hidden_layer_sizes=(10,),
    activation="relu",
    learning_rate_init=0.001,
    max_iter=1000,
    random_state=42
)
```

```
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
```

```
print("Accuracy:", accuracy_score(y_test, y_pred))
```

```
plt.scatter(X[:, 0], X[:, 1], c=y)
plt.title("Multi-Class Data - ReLU, Learning Rate 0.001")
plt.show()
```

IDLE Shell 3.13.5

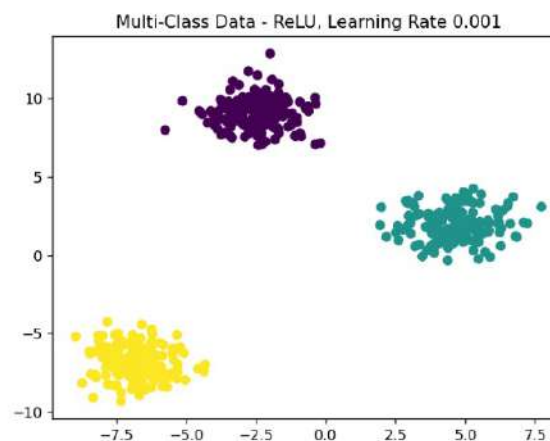
File Edit Shell Debug Options Window Help

Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>

----- RESTART: C:/Users/shaik/OneDrive/Desktop/MLA0405/exp 28.py -----
Accuracy: 1.0

Figure 1



Ln: 5 Col: 0

Ln: 37 Col: 0

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exp 27.py - C:/Users/shaik/OneDrive/Desktop/MLA0405/exp 27.py (3.13.5)

```
File Edit Format Run Options Window Help
import matplotlib.pyplot as plt
from sklearn.datasets import make_classification
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score
```

```
X, y = make_classification(
    n_samples=500, n_features=2, n_redundant=0,
    n_informative=2, n_classes=2, random_state=42
)
```

```
x_train, x_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

```
model = MLPClassifier(
    hidden_layer_sizes=(10,),
    activation="relu",
    learning_rate_init=0.001,
    max_iter=1000,
    random_state=42
)
```

```
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
```

```
print("Accuracy:", accuracy_score(y_test, y_pred))
```

```
plt.scatter(X[:, 0], X[:, 1], c=y)
plt.title("Two-Class Data - ReLU, Learning Rate 0.001")
plt.show()
```

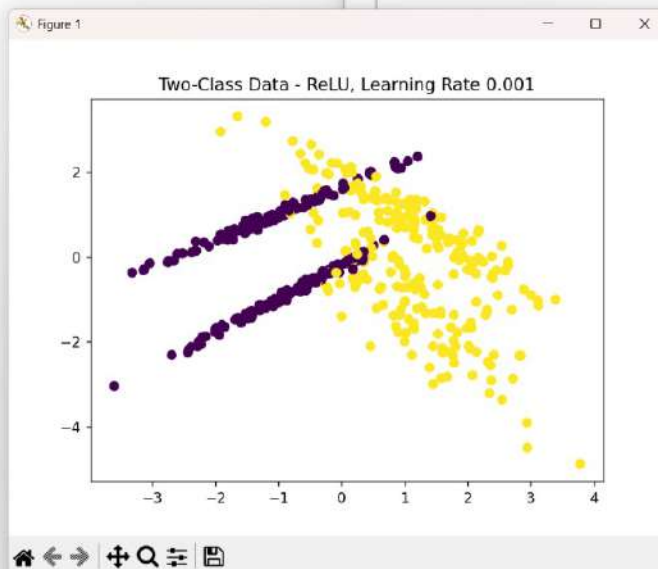
IDLE Shell 3.13.5

File Edit Shell Debug Options Window Help

Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>

===== RESTART: C:/Users/shaik/OneDrive/Desktop/MLA0405/exp 27.py =====
Accuracy: 0.89



```

exp 26.py - C:/Users/shaik/OneDrive/Desktop/MLA0405/exp 26.py (3.13.5)
File Edit Format Run Options Window Help
import matplotlib.pyplot as plt
from sklearn.datasets import make_blobs
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score

X, y = make_blobs(
    n_samples=500, centers=3, n_features=2,
    random_state=42
)

x_train, x_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

model = MLPClassifier(
    hidden_layer_sizes=(10,),
    activation="tanh",
    max_iter=1000,
    random_state=42
)

model.fit(X_train, y_train)
y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))

plt.scatter(X[:, 0], X[:, 1], c=y)
plt.title("Multi-Class Data - Tanh Activation")
plt.show()

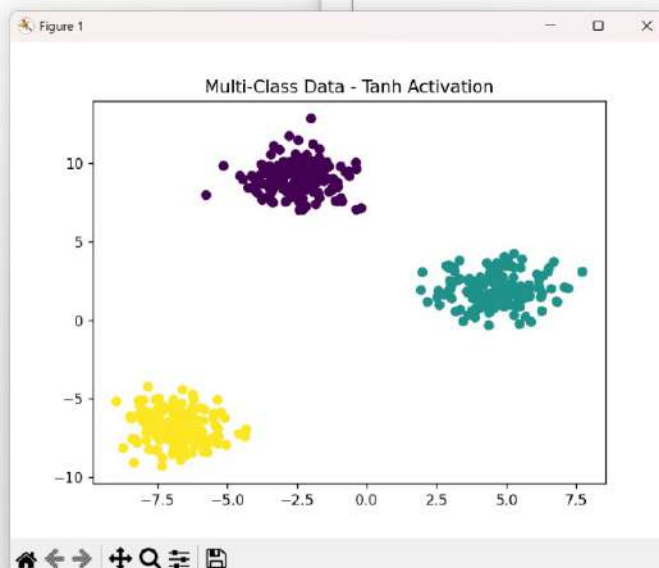
```

```

Python 3.13.5
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/shaik/OneDrive/Desktop/MLA0405/exp 26.py =====
Accuracy: 1.0

```



exp 22.py - C:/Users/shaik/OneDrive/Desktop/MLA0405/exp 22.py (3.13.5)

```
File Edit Format Run Options Window Help
import matplotlib.pyplot as plt
from sklearn.datasets import make_classification
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score

X, y = make_classification(
    n_samples=500, n_features=2, n_redundant=0,
    n_informative=2, n_classes=2, random_state=42
)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

model = MLPClassifier(
    hidden_layer_sizes=(10,),
    activation="relu",
    max_iter=1000,
    random_state=42
)

model.fit(X_train, y_train)
y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))

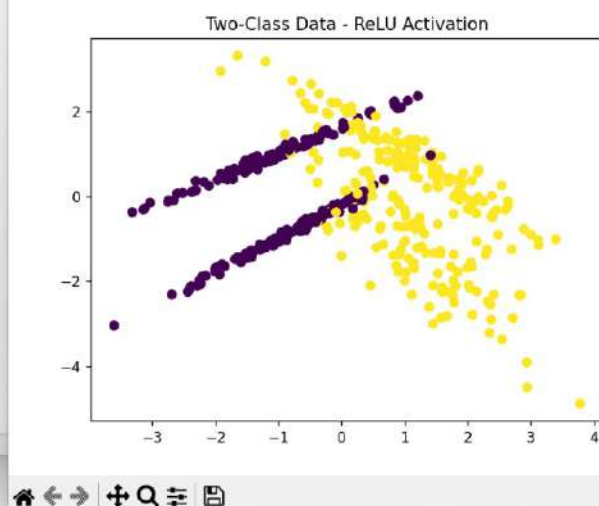
plt.scatter(X[:, 0], X[:, 1], c=y)
plt.title("Two-Class Data - ReLU Activation")
plt.show()
```

IDLE Shell 3.13.5

```
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Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/shaik/OneDrive/Desktop/MLA0405/exp 22.py =====
Accuracy: 0.89
```

Figure 1



Ln: 36 Col: 0

exp 24.py - C:/Users/shaik/OneDrive/Desktop/MLA0405/exp 24.py (3.13.5)

File Edit Format Run Options Window Help

```
import matplotlib.pyplot as plt
from sklearn.datasets import make_blobs
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score

X, y = make_blobs(
    n_samples=500, centers=3, n_features=2,
    cluster_std=1.5, random_state=42
)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

model = MLPClassifier(
    hidden_layer_sizes=(10,),
    activation="logistic",
    max_iter=1000,
    random_state=42
)

model.fit(X_train, y_train)
y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))

plt.scatter(X[:, 0], X[:, 1], c=y)
plt.title("Multi-Class Data - Sigmoid Activation")
plt.show()
```

Ln 36 Col 0

"IDLE Shell 3.13.5"

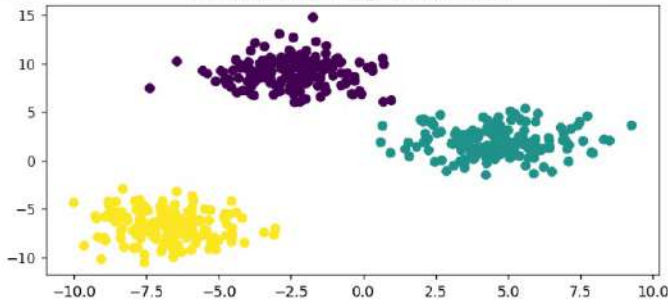
File Edit Shell Debug Options Window Help

Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/shaik/OneDrive/Desktop/MLA0405/exp 24.py =====
Accuracy: 1.0

Ln 5 Col 0

Figure 1

Multi-Class Data - Sigmoid Activation



The scatter plot displays three distinct clusters of data points in a 2D space. The x-axis is labeled from -10.0 to 10.0 with major ticks every 2.5 units. The y-axis is labeled from -10 to 15 with major ticks every 5 units. The first cluster, colored yellow, is located in the lower-left region with x-values between -10 and -5 and y-values between -10 and -5. The second cluster, colored purple, is located in the upper-left region with x-values between -10 and -2 and y-values between 5 and 15. The third cluster, colored teal, is located in the middle-right region with x-values between 0 and 10 and y-values between 0 and 5.

Ln 5 Col 0

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